

Team 15: Dynamic Adjustable RC Car

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Internal Design Review v. 2.0

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Introduction

In many mechanical engineering courses, there is an emphasis on theoretical analysis, but students often lack the hands-on tools to visualize and validate these concepts. This disconnect was identified as Dr. Carlos Hidrovo prepared for his upcoming mechanical engineering course, *Fundamentals of High-Performance Road Vehicle Dynamics* (ME 4690).

Existing commercially available radio-controlled (RC) cars are not designed for controlled experimentation. Their fixed geometry and limited instrumentation restrict the study of how changes in the car's parameters, such as center of gravity, wheelbase, and track width, impact its performance. This leaves this option unsuitable for use as an instructional tool. To address this problem, Team 15 is designing and building an RC car with adjustable center of gravity, wheelbase, and track width, along with sensors to measure the vehicle's response. This system will support controlled demonstrations that link configuration changes with observable dynamic behavior.

Literature Review

1. Overview

This section summarizes research supporting the design decisions for an RC vehicle with adjustable geometry and CG. It starts by reviewing key vehicle dynamics principles, such as load transfer, understeer/oversteer, and CG location. These principles establish the theoretical basis for the design. Next, the section explores localization technologies to determine accurate measurement of the car's motion. Additionally, existing RC car platforms are examined to highlight their limitations, primarily their lack of adjustability. This research shows a gap between existing solutions and experimental needs, justifying the product's development.

2. Relevant Theory

1. Load Transfer

A fundamental aspect of vehicle dynamics, and this project is load transfer, which refers to the redistribution of normal forces among the tires during acceleration, braking, and cornering. Although the total vehicle weight remains constant, inertial forces acting through the center of gravity create moments that shift the load between axles and between the sides of the vehicle. The textbook *Race Car Vehicle Dynamics* [1] explains that longitudinal load transfer during acceleration or braking can be expressed as:

$$\Delta W = (h * W * ax)/L \quad (1)$$

where h is the CG height, W is vehicle weight, ax is longitudinal acceleration, and L is wheelbase. Similarly, lateral load transfer during cornering is approximated by:

$$LLT = ay * h/t \quad (2)$$

where ay is lateral acceleration and t is track width.

These relationships demonstrate that CG height, wheelbase, and track width directly influence load distribution across the tires. Because tire force generation is nonlinear with respect to vertical load, excessive load transfer reduces total available traction and affects handling balance.

2. Understeer and Oversteer

Understeer occurs when the car's turning angle is less than the steering angle, while oversteer occurs when it is greater. To analyze these phenomena, the understeer gradient (UG) is used. The equation for UG is:

$$UG = \frac{d\delta}{dA_y} (3)$$

where $d\delta$ is the change in the steering angle from the Ackermann angle, and dA_y is the change in the lateral acceleration. The Ackermann angle accounts for the fact that the inner and outer front wheels must turn at different angles when cornering. Deviations from this ideal condition contribute to understeer or oversteer, depending on how the front and rear tires reach their grip limits.

3. Center of Gravity

Accurate CG measurement is essential for evaluating the effects of changes in mass distribution. Analytical calculation of CG location using integration is impractical for complex assemblies composed of multiple materials and irregular shapes. Experimental techniques such as suspension methods can be used, in which the object is hung from multiple points and vertical lines are traced to determine the intersection that represents the CG [2], [3].

Alternatively, computer-aided design (CAD) software can estimate CG by assigning material properties to modeled components and numerically calculating the mass distribution [4]. The accuracy of this approach depends on how closely the digital model matches the physical system [5].

This semester, the team decided to use a different experimental method of CG determination, provided by *Race Car Vehicle Dynamics* [1]. Determining the horizontal location involved placing the car on four scales to determine the individual wheel loads. The results from this scaling can be turned into a CG location

$$b = \frac{W_f l}{W} (4)$$

where b is the distance from the front axle to the CG, W_f is the total load on the front wheels, l is the wheelbase, and W is the total weight of the car. For the CG height, the car must be lifted by its rear axle at an angle θ . The resulting height is:

$$h = R + \frac{W_f l - Wb}{W \tan \theta} (5)$$

where R is the tire radius.

3. Telemetry/Electronics

For this project, accurate estimation of vehicle position and orientation is necessary to correlate geometric changes with performance metrics, which can be achieved through sensor fusion techniques that combine inertial measurement unit (IMU) data with global navigation satellite system (GNSS) measurements or visual odometry [6]. IMUs such as the BNO055 and BNO085 integrate accelerometers, gyroscopes, and magnetometers with onboard sensor fusion algorithms to estimate orientation and motion [6]–[8]; while these sensors provide accurate short-term motion estimation, they are susceptible to drift over time and therefore require correction using absolute positioning data such as GPS [9]. This sensor fusion is implemented using either a complementary filter or, for more accuracy, an Extended Kalman Filter (EKF). [7]

Additional sensors, such as wheel encoders, further enhance localization accuracy by enabling wheel odometry and slip detection. The integration of these measurement technologies enables the proposed platform to collect quantitative, research-grade data necessary to evaluate vehicle dynamics.

4. Existing Solutions

RC car manufacturers such as Traxxas, Arrma, and Losi offer a wide range of configurations. *Traxxas* sells drivetrains with two-wheel and four-wheel options, different motor types (brushed or brushless), and scale [10], [11]. Most vehicles fall between a 1/18 and 1/7 scale and allow for minor changes in ride height, but the parameters in this project are fixed. As a result, modifying the key characteristics (track, wheelbase, and CG) requires purchasing a car of a different size.

Initial Designs

1. Center of Mass Iterations

1. Discrete Grid Method

This method involves creating or modifying the base chassis unit to a design that includes several different holes/slots at discrete positions (i.e., 20 centimeter increments away from the nominal COM) and placing custom-made masses into the slots to offset the COM.

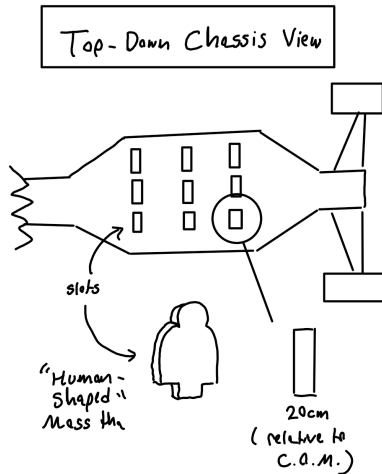


Figure 1: Sketch of a potential layout for the *Discrete Grid* method

Along with the initial sketches and brainstorming of this idea, a preliminary CAD prototype was made to test its efficacy and how it interacted with the existing mounting points found on our purchased Traxxas model.

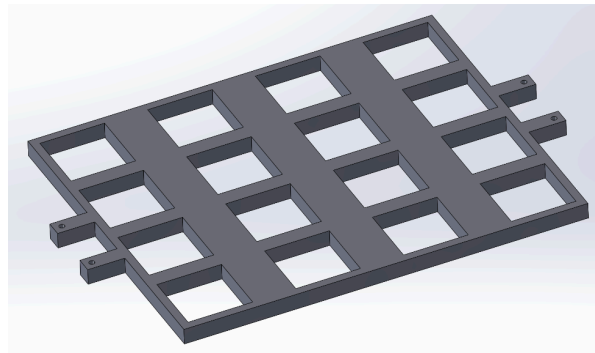


Figure 2: Preliminary prototype CAD model of the *Discrete Grid* method

The positives of this design: the design is the easiest to curate and implement. The robustness and ease of mounting the mass in one place without it moving or separating while the car is moving. Ideally, the masses would snap-fit or use a poka-yoke method to connect to the grid. Some negatives of this design: the inability to place the mass at any position we want due to the discrete positions. We would also have to machine our own custom masses, which adds production time to our workflow. Additionally, this method would have to telescope and adjust its length as the car's wheelbase increases or decreases, adding complexity to the design and affecting the distances between discrete positions.

2. Rack & Pinion / Telescoping Guide Rail Method

This method involves a rack-and-pinion-esque mechanism mounted along the length of the car, with a weight or mass attached to the rack. The longitudinal component of the COM would change as you slide the entire pinion along the car's length, and the vertical/normal component would change by using a crank to move the internal rack along the pinion.

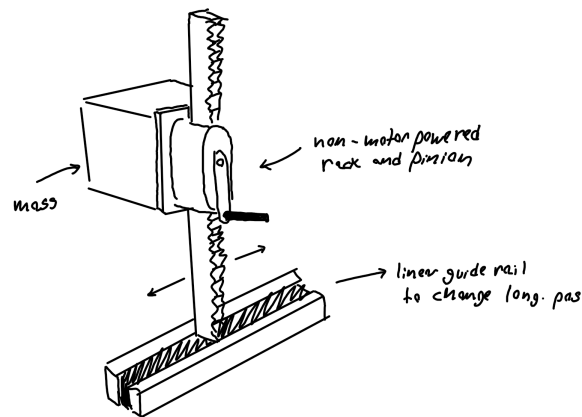


Figure 3: Sketch of Linear Guide Rail

A very similar idea was created with the intention of utilizing 80/20 t-slots (an off-the-shelf construction system made of aluminum bars that interlock, akin to a child's Erector Set) that come with linear bearings that slide along a machined slot in the bar. These linear bearings would house the different masses and come with a brake that allows you to stop the bearing at any desired position along the rail. The aforementioned design is pictured in Figures 3 and 4.

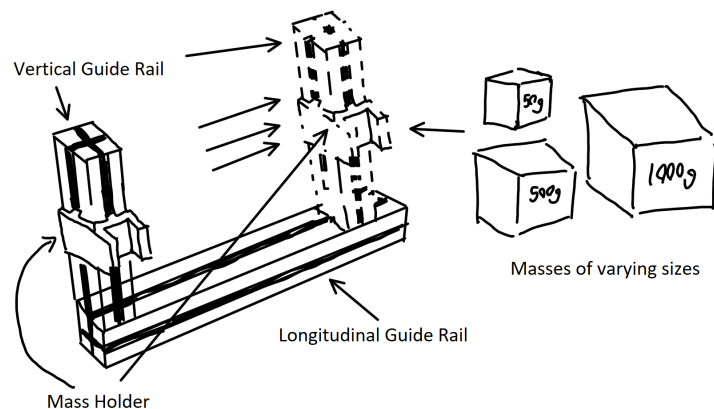


Figure 4: Lateral Mass Component

Not pictured in the above sketches is a component of the mechanism that maneuvers the mass laterally along the width of the car. That component is shown in Figure 5 below, which depicts a preliminary CAD model of a prototype of the *Telescoping Guide Rail* method.

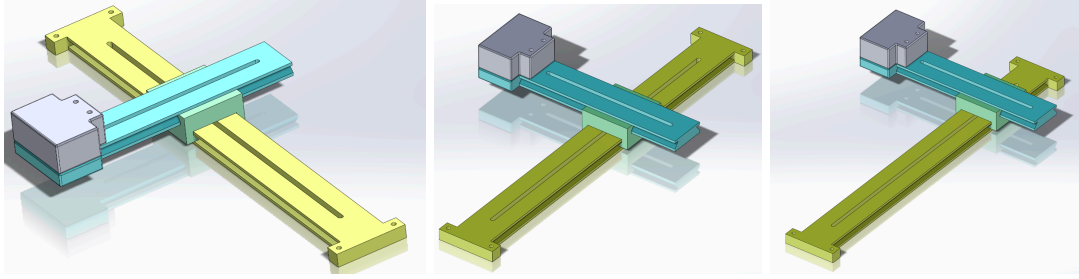


Figure 5: CAD of Telescoping Guide Rail

The positives and negatives of both designs are virtually identical. With this design, we would gain the ability to place the mass at any position we want, with the added benefit of being able to create markings on the rail to make our own equally spaced discrete positioning. Additionally, this method would be relatively easy to implement because we can have the OTS parts cut to length from the company that provides them, if needed. This design has the potential to come loose as the vehicle is in motion due to vibrations because the OTS brake has to be manually cranked by hand, so it does not have the capacity to be tightened with excessive force so as not to impede the user experience with the mechanism. Like the *Discrete Grid* method, the rail that goes along the length of the vehicle would need to telescope as the wheelbase is varied, adding increased complexity.

2. Track Iterations

1. Wheel Spacers

This method involves creating spacers between the wheels of the vehicle to vary the track by moving the wheels farther away from the center of the chassis. The spacers (for the time being) would be 3D-printed in various lengths so that they can be swapped out as needed.

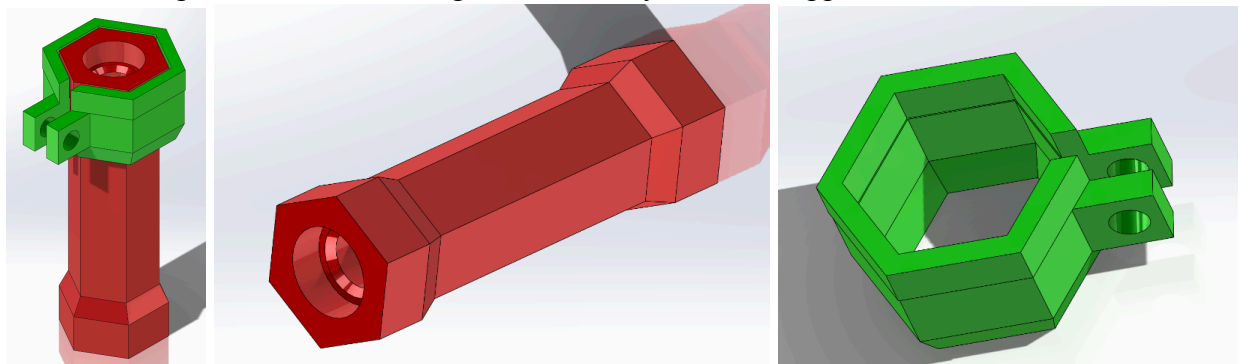


Figure 6: Images of CAD model for a preliminary design of the *Wheel Spacer* method used for testing

This design is simple to implement and fabricate at scale, with the printing of this component in PLA (Polylactic Acid) only taking roughly an hour. Adversely, quick manufacturing using 3D prints increases the risk of the spacers breaking apart from forces experienced while the car is in

motion. This could be prevented by being careful during testing and limiting the speed at which we drive, which would inherently limit our data and the full range of testing. We are considering using 3D prints as intermediaries and moving to machined versions for future or final designs.

Below is a diagram of the purchased Traxxas Rustler 4x4 with the internals exposed to help visualize where the above designs will be mounted or implemented.

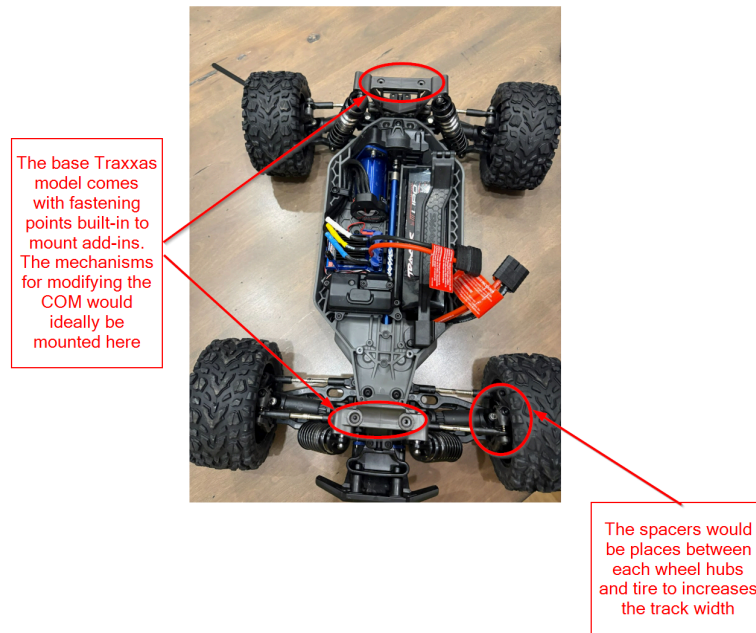


Figure 7: Diagram detailing mounting solutions on the Traxxas Rustler 4x4

Decision Criteria

1. Structural Components

There were two structural design considerations for both the wheelbase and center of gravity that were constructed and compared in a Pugh matrix for a better understanding of which may be the better fit. These matrices are shown in Tables 1 (for the wheelbase) and 2 (for the CG) below. The team decided to use time, feasibility, cost, effectiveness, long-term benefit, user experience, and customizability as the categories for this decision-making process.

Table 1: Pugh Matrix Comparison for Wheelbase Design Options

		Options			
Criteria	Weighting	Design 1: Swappable Chassis Components		Design 2: Telescoping Chassis	
		Score	Total	Score	Total
Time	3	4	12	2	6

Feasibility	4	3	12	3	12
Cost	3	2	6	3	9
Effectiveness	5	2	10	4	20
Long-term benefit	2	3	6	3	6
User Experience	5	1	5	4	20
Customizability	3	2	6	5	15
	Totals		57		88

Table 2: Pugh Matrix Comparison for Center of Gravity Design Options

Criteria	Weighting	Options			
		Design 1: 80/20 Linear Guide Rail		Design 2: Grid-Based System	
		Score	Total	Score	Total
Time	3	3	9	4	12
Feasibility	4	4	16	4	16
Cost	3	3	9	4	12
Effectiveness	5	4	20	3	15
Long-term benefit	2	3	6	3	6
User Experience	5	4	20	3	15
Customizability	3	5	15	2	6
	Totals		95		82

After this analysis, the telescoping chassis was the better wheelbase design, and the linear guide rail was the better CG design. Both designs for the center-of-gravity considerations were physically fabricated via 3D printing to enable a more accurate comparison of their practicalities. By observing the functionality of the results and considering Professor Hidrovo's insight, our final design decisions support our original Pugh matrix results and conclude that the telescoping chassis and linear guide rail shown in Figure 5 are the final design choices.

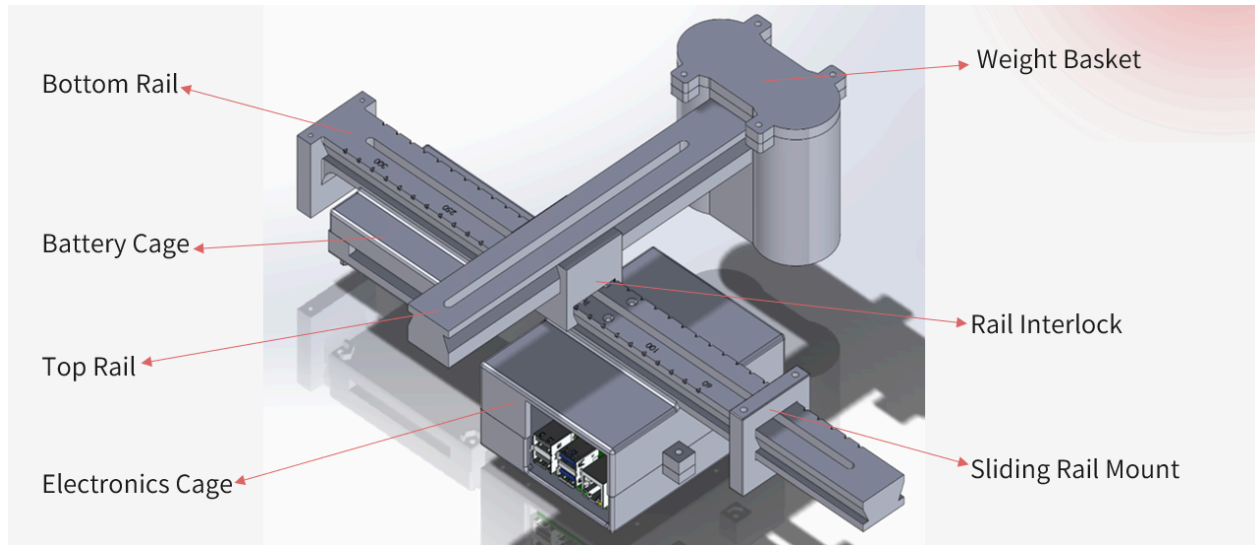


Figure 8: Final Linear Guide Rail System

Figure X shows the final design of the linear guidewheel system, which encompasses the electronics (and their enclosures), extra battery, and linear rail system. The central rail is mounted across the length of the Traxxas unit, while the upper rail slides along the length of the unit. The upper rail is free to move along the width of the car and has a modular weight basket to place calibration weights. There are various weight baskets of different sizes corresponding to different calibration weights. This design was selected because it provides additional space for components that could not fit on the original chassis plate. It also allows the center of gravity to shift along the x-plane, both front-to-back and laterally across the chassis, offering maximum flexibility in adjusting mass distribution. While retaining a strong safety factor of 4 and a max displacement on the bucket of 1.5mm, as shown by the simulations on the complete system in the figure below.

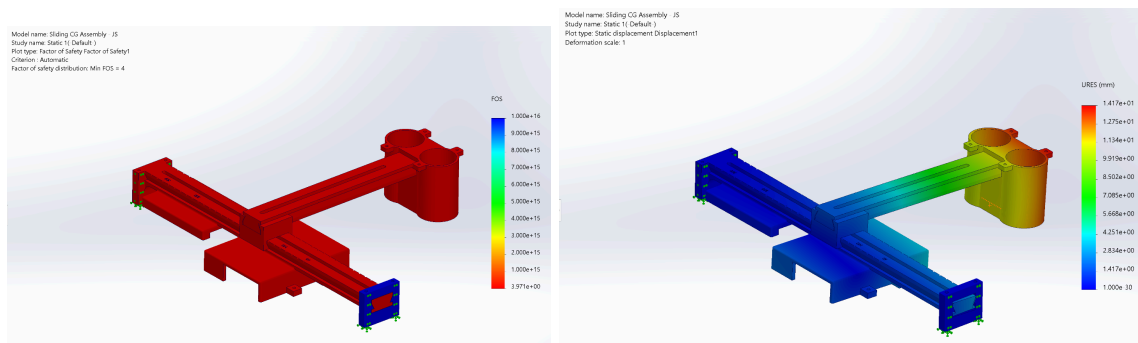


Figure 9: Final Linear Guide Rail System Max Load Test (1kg)

As for the custom wheel spacers, we retained the original design from the initial concepts. However, after Phase 2 of mechanical testing, the print orientation was adjusted to improve structural strength and provide a more secure and reliable mechanism. Below is the wheel spacer design we incorporated into the final prototype.

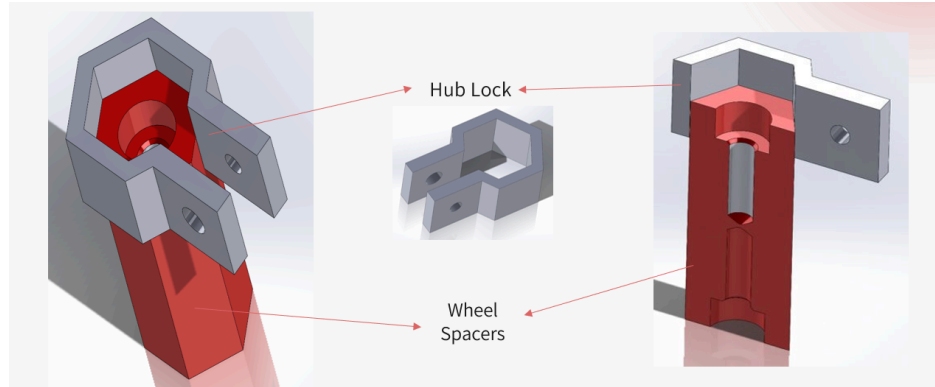


Figure 10: Wheel Spacers

This new design was given an extreme safety factor of 5 for static loading, so as to resist the cyclic loading it will face. It received a minimal displacement of 0.273mm. Both are seen in the following figure. The load was based on the max output of the electric motor, and the end of the spacer was fixed. An extreme condition that would not be experienced while in normal operation.

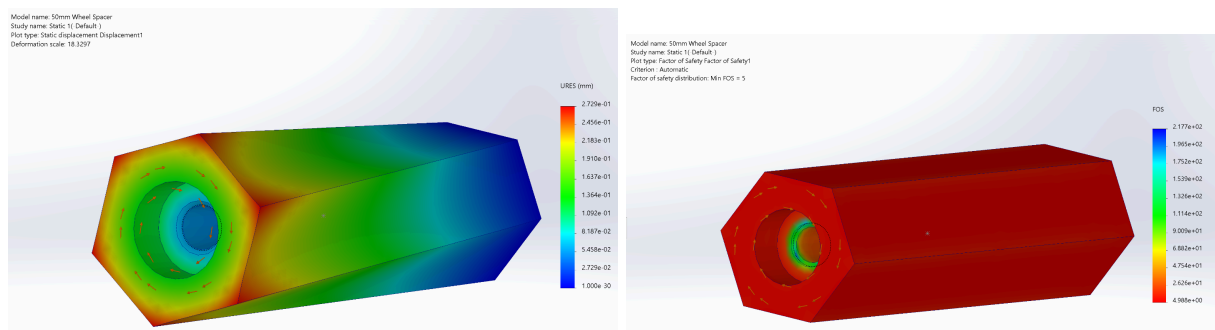


Figure 11: Wheel Spacers Max Load Test (1.5Nm)

With the combination of both designs and our electronics setup, we were able to achieve a working minimum viable prototype that could alter key variables: center of gravity and track. We were able to instrument the car with electronics to measure velocities, accelerations, rotations, and steering angles as a function of the aforementioned key variables. Below is a figure of the completed prototype that was presented during Capstone Day.

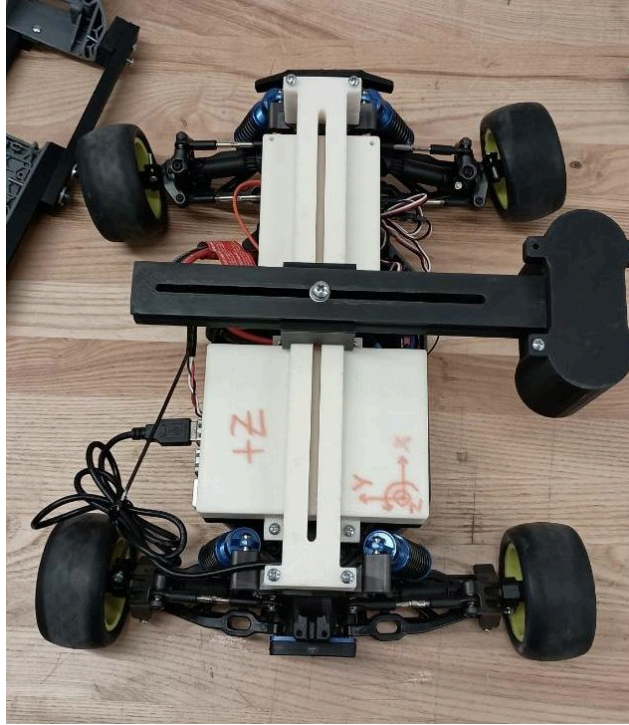


Figure 12: Final Design

Analysis & Testing

This section is structured around four key phases that were conducted to successfully develop the final prototype. Phase One involved initial baseline testing of the car to understand its unmodified performance. Phase Two focused on mechanical testing to evaluate and refine structural components. Phase Three consisted of electronic testing, including code calibration and sensor validation. Finally, Phase Four involved comprehensive testing of the fully integrated system, combining all mechanical and electrical components to assess overall performance.

Phase I: Preliminary Testing of Traxxas Car without Modifications

On February 15th, the newly ordered Traxxas car was tested prior to any mechanical or electrical modifications to establish a baseline for its performance. This initial testing allowed the team to gauge a maximum speed of approximately 60 mph, as well as observe its handling, acceleration, braking response, and overall stability under normal driving conditions. These baseline measurements provided a reference for evaluating the effects of subsequent modifications.

Foundational center of gravity calculations were also performed using the hanging method described in the literature review. This method determines the height of the center of gravity above the axle centerline using the equation:

$$h_1 = (W_f \cdot \ell - W \cdot b) / (W \cdot \tan \theta) \quad (6)$$

where W_f is the weight on the front wheels with the rear elevated, W is the total vehicle weight, ℓ is the wheelbase, b is the horizontal distance from the rear axle to the CG as determined through the testing outlined in the literature review, and θ is the angle of inclination. Since the vehicle is elevated at the rear during testing, R_L was measured directly and used to find the total CG height using:

$$h = R_{L,CG} + h_1 \quad (7)$$

where R_L , CG is the loaded tire radius, and h is the total height of the CG above the ground.

Negligible tire deformation was observed during testing, so the tire radius was treated as constant across all measurements. The angle of inclination θ was derived from the measured rear tire height using the Pythagorean theorem, with the rear tire height and wheelbase serving as the known values. This process was repeated at three different angles to observe CG trends and ultimately determine the vehicle's center of gravity. The corresponding calculations are shown below.

Preliminary Calculations of Car

Table 4: Horizontal CG Variables for Calculations

W (g)	W _f (g)	W _R (g)	ℓ (in)	d (in)	b (in)	a (in)	y' (in)	y'' (in)	R _L (in)	Front Tire Height
2667	1229	1438	12	0	5.52	6.28	6.007	0.007	2.17	2.8

Table 5: Center of Gravity Calculations

Rear Tire Height	Theta (radians)	Front Left	Front Right	Total Wf	h1	h	AVERAGE
10.7	0.7186024381	658	651	1309	0.4227812735	2.592781273	2.61060587
13.1	1.032012516	618	645	1263	0.09731201388	2.267312014	
8.5	0.4949640317	640	683	1323	0.8017243152	2.971724315	

Table 6: Lateral Load Transfer Calculations

Ay (g's)	h	t	LLT	Total Weight (g)
0.5	2.610605867	12	0.1087752445	290.103577
1	2.610605867	12	0.217550489	580.2071541
1.5	2.610605867	12	0.3263257334	870.3107311
2	2.610605867	12	0.4351009779	1160.414308
2.5	2.610605867	12	0.5438762224	1450.517885
3	2.610605867	12	0.6526514669	1740.621462

Phase II: Mechanical Testing

The first mechanical test was conducted on March 31st to evaluate whether the linear guide rail and wheel spacers were properly installed and capable of withstanding driving conditions. Testing took place outside Snell Engineering, where multiple component failures occurred. As a result, the mechanical design was revised and reprinted to improve structural integrity. These findings provided critical insight into weak points in the system and informed design improvements, enabling the team to move forward with a more robust and reliable setup for future testing.

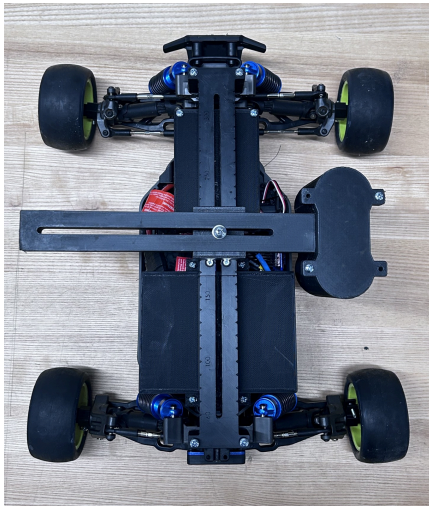


Figure 13: Initial Mechanical Design

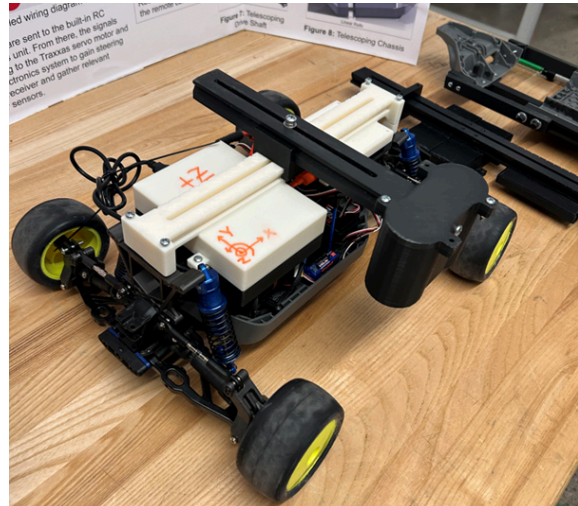


Figure 14: Final Mechanical Design

After finalizing the car's structural components, the team proceeded to Phase Three of testing, which focused on the electrical components.

Phase III: Electrical Testing & Data Collection

The electronics used to collect data on the vehicles' linear acceleration, angular velocity, orientation, and steering angle, as well as the derived states such as linear velocity and position, consisted, at its center, of a Raspberry Pi 4B. This is a single-board computer (SBC) that runs the main controller Python script (*Controller.py*), which handles the computation, logic, and logging required for data collection, and interfaces with the sensors via GPIO pins and USB ports. This board receives 5V power via a USB-C port connected to a portable battery core. Placed on top of the Raspberry Pi 4B is an Adafruit Ultimate GPS hat, which provides latitude and longitude via GNSS and also acts as a GPIO expansion board to distribute signals to system components. This GPS Hat for the Raspberry Pi 4B also acts as a "proto-board", on top of a BNO055 9DOF Absolute Orientation Fusion Inertial Measurement Unit (IMU) is soldered. This sensor, connected to the Raspberry Pi 4B using an Inter-Integrated Circuit (I2C) bus, provides corrected linear acceleration vectors and calculates the sensor's absolute orientation. It does this by implementing proprietary sensor fusion technology that uses Kalman Filters, producing fairly

accurate roll, pitch, and yaw orientation. This sensor fusion combines data from a 3-axis accelerometer, which provides acceleration (including gravity), a 3-axis gyroscope, which provides angular velocity, and a 3-axis magnetometer, which reads the Earth's magnetic field and acts as a compass pointing to the Earth's south pole. By combining the orientation of the sensor derived from the accelerometer and magnetometer data, and the angular velocity in a Kalman filter using sensor fusion, accurate roll, pitch, and yaw orientation Euler values can be found (in reality, these values are read as quaternions, as Euler angles are prone to gimbal lock, but are converted to euler angles for the purposes of logging telemetry).

The Arduino Nano intercepts the PWM signal from the output of the RC Receiver of the vehicle, which is the steering input in parallel to that sent to the 6V waterproof brushless steering servo, making use of the Nano's GPIO pins, allowing the system to extract and measure the steering angle for data collection and analysis without disrupting the original control pathway. By tapping into this signal within the circuit, the Nano acts as an intermediary between the receiver and the rest of the system, reading the PWM pulse width (the duration the PWM signal is at the 3.3V HIGH digital logic level) and mapping it to the vehicle's steering angle. The raw pulse width as well as the mapped steering angle are transmitted to the Raspberry Pi 4B via USB port using serial (UART) communication in 'NMEA' convention at a 11520 baud rate. 'NMEA' is a standardized serial communication protocol to transmit data via ASCII text strings, which implements a checksum for data validity. The checksum is used to verify data integrity, ensuring that the 'NMEA' messages haven't been corrupted during transmission. This enables real-time acquisition of steering input while maintaining normal vehicle operation, allowing the measured data to be integrated with other sensor outputs for a more comprehensive analysis of vehicle dynamics. Combined, the raw data received from the sensors are the linear accelerations in the IMU's body frame, the IMU's absolute orientation, and the mapped steering angle calculated from the pulse width of the PWM signal received by the RC receiver module and output to the vehicle's steering servo.

The controller code [12] running on the Raspberry Pi 4B reads this raw data, performs signal post-processing (since all sensors and electrical components are prone to noise), and calculates the IMU's position and linear velocity via odometry. As the orientation values output by the IMU implement proprietary sensor fusion techniques that utilize Kalman Filters—a recursive algorithm that estimates the state of a dynamic system using a mathematical model of the acceleration-magnetometer readings and the gyroscope's angular velocity—very little post-processing was needed for the IMU's absolute orientation. However, as these orientation readings still encountered occasional, yet rare, peaks, a high-pass filter was applied to ensure accurate readings. The acceleration values were the noisiest, requiring both a simple moving-average filter with a rolling circular buffer of size 3 and a low-pass filter to mitigate high-frequency noise due to vibrations. To calculate the IMU's position and linear velocity, odometry was used to compute acceleration, integrate acceleration to obtain velocity, and integrate velocity to obtain position, using the trapezoidal rule over the measured time step.

However, because this data propagates noise from the acceleration readings, an Extended Kalman Filter (EKF) was implemented to compute accurate position estimates via sensor fusion of GPS and IMU data. This EKF implementation was based on the paper *"Robust Multisensor Localization Using Sparse Landmarks for Autonomous Driving"* by Ellinoudis et al. [7] and on the advice of Professor David Rosen from Northeastern University's Robust Autonomy Lab. The EKF uses a dynamics model and a sensor model to define the probabilistic state-transition and

observation functions. With the assumption of additive Gaussian noise, these induce Gaussian conditional distributions, which are propagated recursively in a Markov framework using predict and update steps. The sensor model, called the noise covariance matrix, for the EKF was determined from the accuracies provided in the spec sheets of the sensors used, as well as from the steering angle, which was measured from the RC Receiver's output ranges. Based on this and a mathematical dynamics model of the system (originally this was attempted as a motion model that incorporated vehicle dynamics, but after discussion with Dr. Hidrovo, this was later changed to an exact implementation of the drone model used in the aforementioned paper). The controller code running on the Raspberry Pi 4B logs telemetry data, including both raw and calculated vehicle state values, and outputs it as a CSV file.

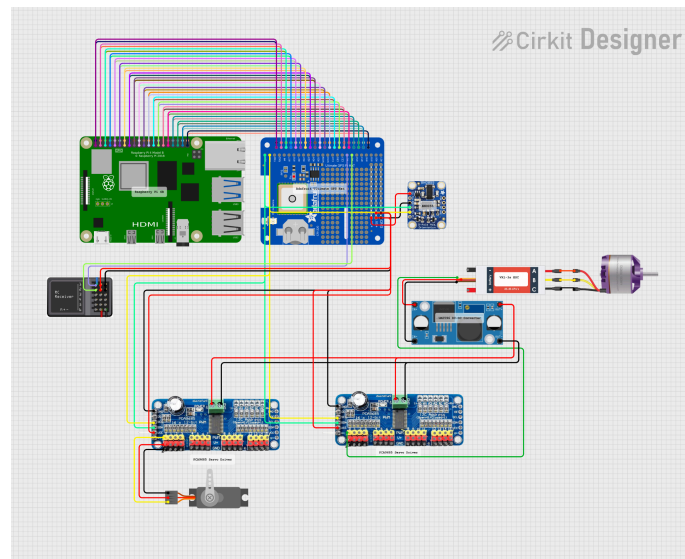


Figure 15: Initial Circuit Wiring Diagram

Additional miscellaneous components include a DC-DC converter that regulates power from the battery and supplies it to the motor driver, servo driver boards, and other electronics, as well as an ADNS1015 Analog to Digital Converter, which is soldered to the GPS hat alongside the IMU, enabling reading analog signals and interfacing with the Raspberry Pi 4B via the I2C bus. Also, a PCA9865 servo driver was initially implemented to control the commanded steering angle to the Raspberry Pi 4B, but this implementation was scrapped in favor of reading the PWM signal in parallel. This choice was due to the determination that noise and latency were unavoidable when reading pulse widths from the RC receiver, and that driving the steering servo would not only affect the vehicle's performance but also propagate noise. The full circuit diagram of the electronics is shown in Figure 15. Our initial circuit design is shown in Figure 16.

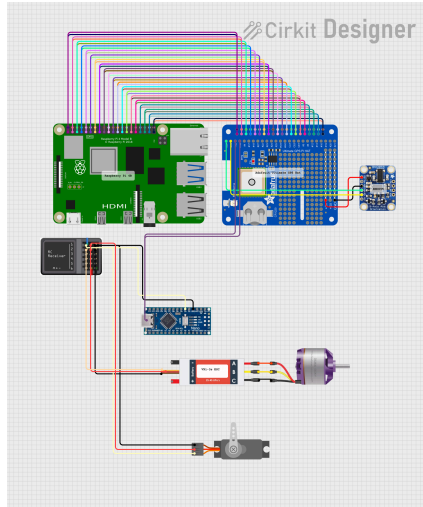


Figure 16: Final Circuit Wiring Diagram

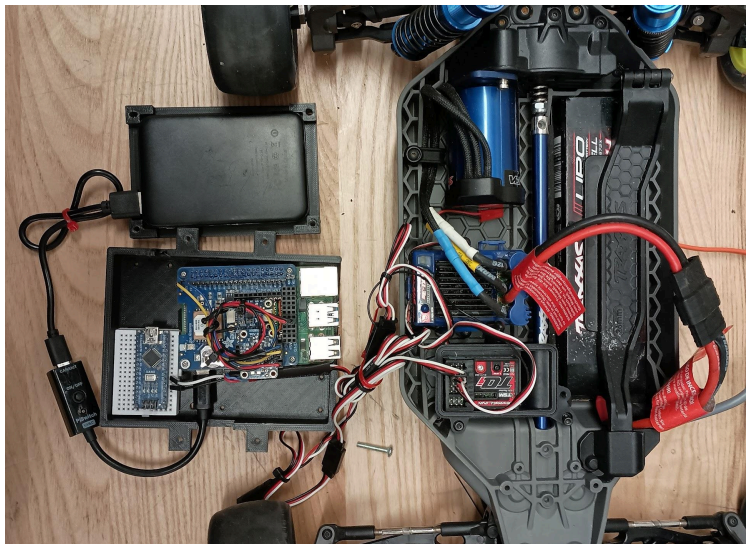


Figure 17: Test Bed Setup

During the electronic testing phase, code calibration and sensor validation were performed to ensure that all components were properly connected and functioning as intended. Prior to integration with the vehicle, sensor testing was performed on a test-bed setup (where the electrical components and sensors were laid on a flat surface and wired together) to verify accurate readings from each sensor individually and to confirm proper wiring and electrical connections. This test bed setup is shown in Figure 17. In addition, initial testing was conducted to validate the controller code and ensure it ran without issues on the Raspberry Pi 4B, and to test the BNO055 IMU's proprietary calibration sequence. This also allowed for determining the necessity of an additional post-calibration sequence for the IMU's acceleration data, as, while the orientation was, again, fairly accurate, the acceleration seemed prone to constant drift. The drift was determined by plotting the acceleration and the calculated velocity and position from the controller code's odometry calculations; it was observed that, while the IMU was held still, the

acceleration drifted with a constant offset, causing the velocity to increase linearly. Subtracting the calculated offset from the acceleration readings allowed the drift in the acceleration readings to be at least temporarily mitigated. In reality, though, the bias of the IMU's acceleration values is neither linear nor time-invariant, but rather an unknown function of temperature. While the IMU's proprietary calibration and sensor fusion mitigated this to a fair degree, and the additional post-calibration helped immensely, these values would eventually drift noticeably and thus require additional calibration.

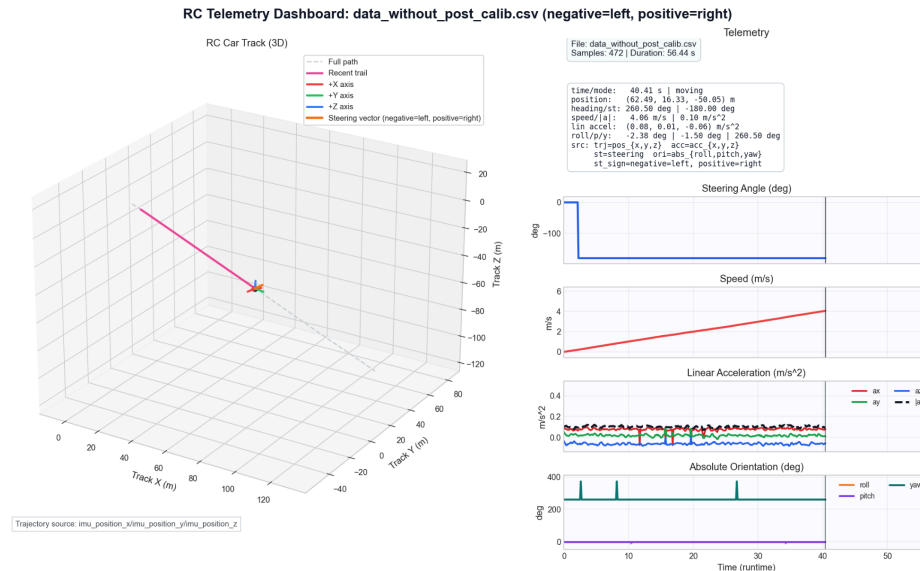


Figure 18: Visualization of the IMU's accelerometer drift while at rest

Once the operation was confirmed successful, the components were installed in the electronic cage of the linear guide rail system and securely mounted to the car. This allowed the team to proceed with final testing under real operating conditions, recording the telemetry output from the controller code as a CSV data file, and visualizing said data using a Python script that plots the values against time (*Plotting.py*), as well as a script that animates the position of the vehicle while simultaneously plotting the steering angle, acceleration, orientation, and velocity (*Animate.py*).

Phase IV: Integrated System Testing

Phase Four represented the testing of the fully integrated system, in which all mechanical and electrical components were installed on the vehicle. During this phase, the car was driven under real operating conditions, allowing the onboard sensors to capture key data such as position, velocity, acceleration, and steering angle. This enabled consistent data collection across each test variation and trial, providing a basis for comparing results and analyzing trends in vehicle performance.

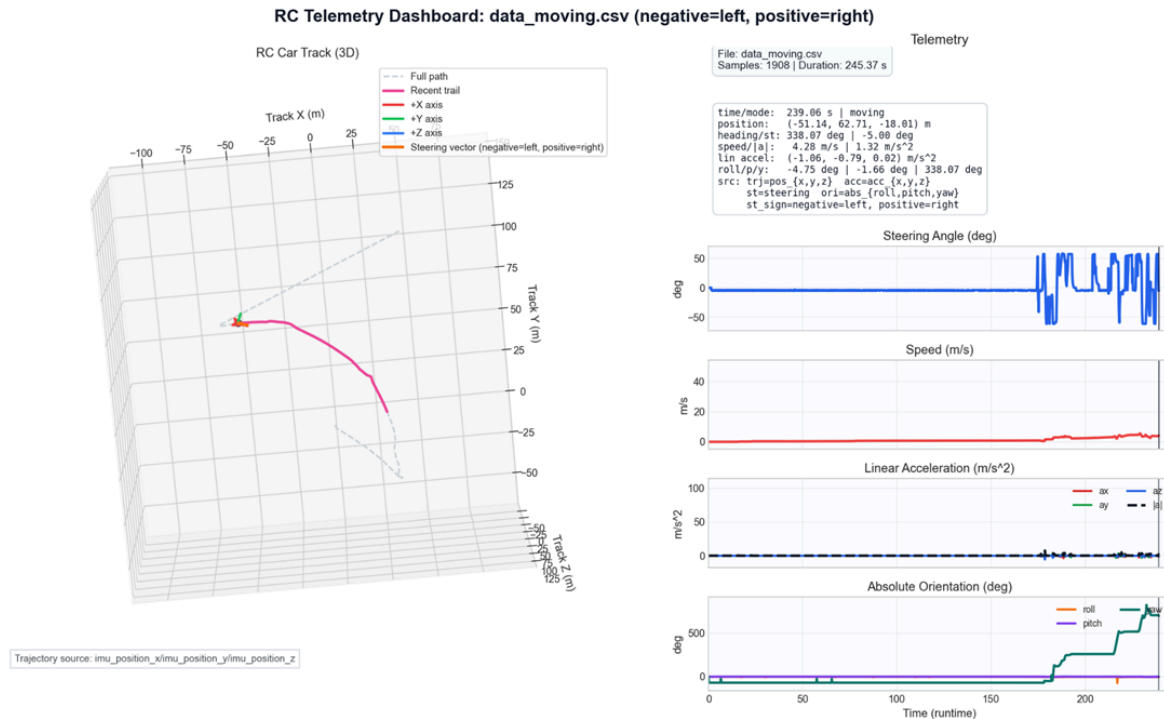


Figure 19: Data Result from Final Test Drive

Figure 18 graphically presents the results obtained from the final test drive, including steering angle, speed, linear acceleration, and axle orientation for a specific center of gravity (CG) and track width combination. For this trial, the configuration included one wheel spacer and a CG positioned approximately halfway from front to rear and centered laterally across the chassis. This process can be repeated for any CG and track width configuration, allowing the results to be compiled and compared. The resulting plots are then used to identify trends across different configurations and support a more in-depth analysis of vehicular dynamics.

Next Steps

The next steps for the project include developing a telescoping chassis paired with a telescoping driveshaft, which will allow controlled adjustment of the vehicle's track length while maintaining a continuous mechanical connection between the front and rear assemblies. This enhancement will expand the system's ability to vary geometric parameters beyond the existing track width and CG adjustments. In addition, an automated braking system can be implemented to provide more reliable, controlled stopping, thereby reducing reliance on passive drivetrain resistance and improving overall vehicle performance and safety. Please see design iterations below:

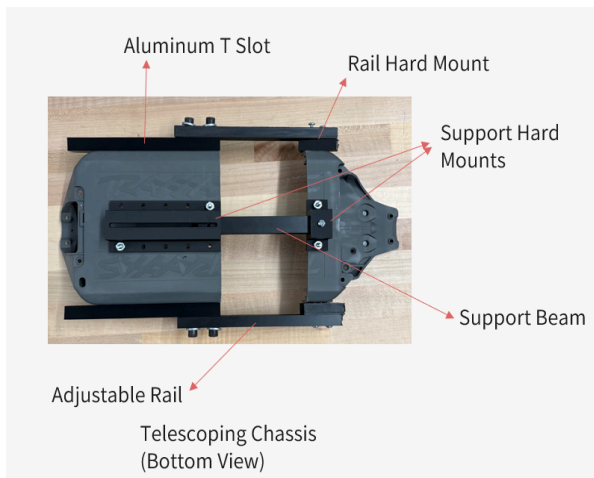


Figure 20: Telescoping Chassis

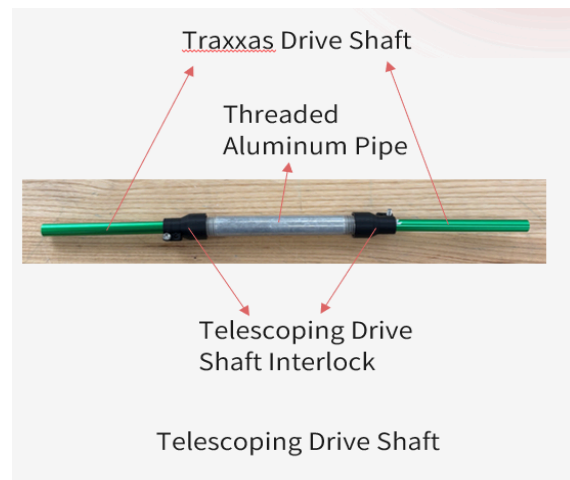


Figure 21: Telescoping Drive Shaft

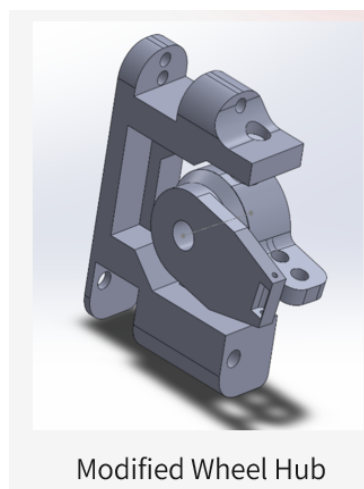


Figure 22: Braking System

Summary

This project successfully led to the development of a fully functional prototype capable of dynamically adjusting both track width and the center of gravity (CG). The implementation of a linear guide rail system, combined with custom-designed wheel spacers, enabled controlled variation of track width while also allowing precise manipulation of the vehicle's mass distribution.

In addition to the mechanical design, the integration of electronic components and sensors was completed successfully. The system captured key performance data, including steering angle, position, velocity, and acceleration. These measurements allowed for the generation and comparison of graphical results across different test configurations, providing valuable insight into the vehicle's dynamic behavior.

Looking ahead, future improvements could include implementing a telescoping chassis and driveshaft system to allow an adjustable wheelbase length in conjunction with existing track width and CG modifications. Additionally, incorporating an automated braking system would enhance vehicle control, as the current platform relies primarily on passive drivetrain resistance rather than an active braking mechanism. These enhancements would further expand the system's capabilities and provide greater flexibility for continued experimentation and analysis.

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